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State-of-the-Art Review

Advancements of Digital Twin Technology for Bridges and Tunnels: A Systematic Literature Review

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Abstract: Bridges and tunnels are essential transportation infrastructure components, yet they are highly susceptible to deterioration due to natural aging and unexpected factors. As a result, continuous monitoring and proactive maintenance are essential to ensure their safety and functionality. The integration of advanced technological solutions into asset management is becoming increasingly important for companies to stay competitive. Digital twins have emerged as a promising tool to optimize asset performance and enhance decision-making. Consequently, advancing and updating knowledge on this technology is of paramount importance. This paper presents a systematic literature review that explores the current state of digital twin technology in civil infrastructure, with a particular focus on bridges and tunnels. Drawing on data from 89 relevant research papers, it contributes to the field by examining the frequency of digital twin applications across various life-cycle phases, evaluating their specific purposes and uses, identifying the most commonly applied technologies for these critical infrastructures, and assessing the level of development based on the integration of technologies. Additionally, the paper highlights key challenges and provides directions for future research regarding the application of digital twins to bridges and tunnels. **DOI: 10.1061/JITSE4.ISENG-2745.** © *2025 American Society of Civil Engineers.*

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Introduction

Civil infrastructure is a significant part of any nation's critical assets and is indispensable for economic development. Its suboptimal performance or performance that is lower than expected can severely impact the country's gross domestic product (Aktan et al. 1998). Bridges and tunnels are critical components of modern infrastructure, ensuring connectivity and supporting economic activities. The safety and functionality of these structures are paramount, as their failure can lead to catastrophic consequences, including loss of life, economic disruption, and extensive repair costs.

Bridges and tunnels exhibit distinct characteristics and challenges that distinguish them from other infrastructure systems. Bridges face challenges related to dynamic load-bearing, material degradation, and environmental exposure, requiring careful monitoring and maintenance. Conversely, tunnels are characterized by their geotechnical complexity, confined space, water infiltration, ventilation, and emergency safety considerations. Hence, ensuring the proper functioning of these infrastructures requires implementing effective preventive and predictive maintenance practices. These practices are essential for minimizing deterioration effects, reducing the risk of failure, minimizing costly repairs, and extending the infrastructure's operational lifespan.

The implementation of smart technologies can effectively address infrastructure challenges and meet growing demands (Berglund et al. 2020). One such emerging technology, the digital twin (DT), is gaining increasing attention as an integrative tool to improve asset performance via advanced computational techniques (Wang et al. 2022). A DT is a dynamic, realistic digital representation of a physical product, system, or process and is continuously updated with real-time data (Mahmoodian et al. 2022). It represents a significant evolution in modeling by extending traditional static information with dynamic data, including both data and algorithms that help explain and predict the behavior of the physical system. By leveraging sophisticated physical models and asset information, a DT mirrors the life of its real-world counterpart (Dan et al. 2022; Glaessgen and Stargel 2012). This connection between physical and virtual environments enables operation management throughout the asset life cycle, facilitating real-time monitoring and control of its operations (Gürdür Broo et al. 2022).

The level of development of a digital twin is closely related to technological advancements and the integration of new technologies into the virtualization process. While recent efforts have been made to characterize and standardize digital twins (Boyes and Watson 2022; Jones et al. 2020), clear standards and definitions for their creation and implementation are still lacking. This lack of standardization leads to ambiguity and makes distinguishing between a digital twin and traditional virtual models challenging. Therefore, well-established methodologies, such as building information modeling (BIM), bridge information modeling (BrIM), and finite-element modeling (FEM), are essential components for the development and evolution of digital twins, serving as key reference points and foundational tools.

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The applications of DTs are diverse, including smart cities, healthcare, automotive, aerospace, and construction industries (Fuller et al. 2020). The adoption of digital twins holds significant potential for improving infrastructure management and enhancing operational efficiency. However, the civil engineering industry is often slower in adopting new technologies, which can introduce environmental, economic, and human risks. Therefore, it is fundamental to explore and disseminate updated knowledge on this technology to facilitate its broader adoption in civil engineering practices.

While previous research has reviewed trends and applications of DTs in infrastructure (Taherkhani et al. 2024), the primary technologies involved (Long et al. 2024), data integration, interoperability, and technological adaptation (Michael et al. 2024; Larbi et al. 2025), workflow and applications of digital twins (Pregnolato et al. 2022), and an industrial overview on digital twin definitions, practices, benefits, and challenges (Broo and Schooling 2023), this study makes a unique contribution by providing a comprehensive and up-to-date systematic literature review with a specific focus on bridges and tunnels, critical assets in transportation infrastructure. This paper advances the state-of-the-art by exploring the specific life-cycle phases in which digital twins are applied to bridges and tunnels, investigating their intended purposes and the key technologies involved and evaluating the current stage of development for digital twins in each life-cycle phase. Furthermore, the review identifies the main challenges and future perspectives on the adoption of digital twins in critical infrastructures such as bridges and tunnels.

Methodology

In the context of this paper, the systematic literature review (SLR) development was fragmented into three key blocks of chronologically ordered activities: planning, execution, and summarization (Fig. 1), as established by Kitchenham et al. (2015). The planning stage is characterized by SLR protocol elaboration, in which the research question (RQs) and search strings related to the topic of interest are defined. The research databases, inclusion/exclusion criteria, and quality assessment metrics are also determined in this phase.

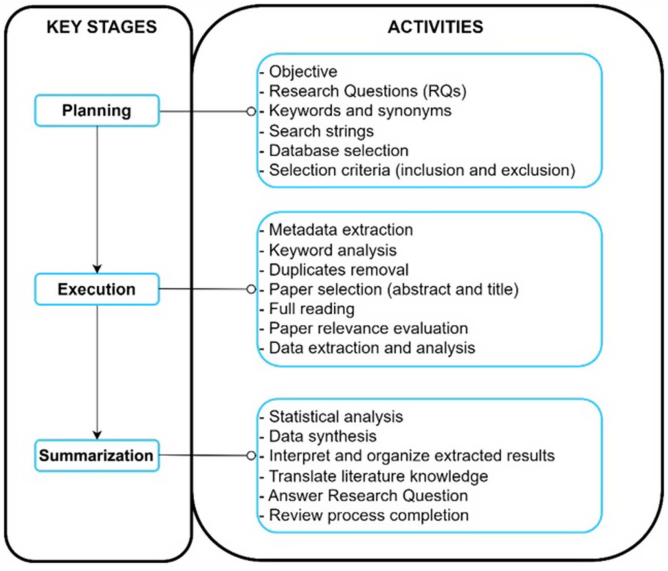


Fig. 1. Stages and activities to develop the SLR.

In an effort to bolster the construction and employment of the review protocol, a framework was developed using the population, intervention, comparison, outcome, and context (PICOC) criterion (as illustrated in Table 1). This criterion was employed to elucidate the target population for evidence extraction, discuss the specific intervention incorporated in the study, establish the comparative control against which the intervention was appraised, ascertain the resulting outcomes, and provide an understanding of the context or practical significance (application) of these findings (Wohlin et al. 2012).

To address the literature gaps in the use of digital twins for bridges and tunnels, four RQs are established as follows:

- RQ1: During which specific life-cycle phases of civil infrastructure assets are digital twins most commonly utilized, and with what frequency are they applied to bridges and tunnels?
- RQ2: What specific purposes underlie the implementation of digital twins in civil infrastructure, such as bridges and tunnels?
- RQ3: What are the associated technologies and levels of development of DT for bridges and tunnels?
- RQ4: What are the primary challenges associated with the adoption of digital twins in civil infrastructure, and which future perspectives can be identified for advancing their utilization in the field?

Trial keyword searches revealed that Scopus, Engineering Village, and Springer citation databases yielded consistent results. Therefore, Scopus, Engineering Village, and Springer were selected as the citation databases for this paper. Digital twins constitute a novel and broad field of study with a diverse scope of utilization, crossing many fields of study. To ensure that all the relevant literature was captured, data extraction from publications was developed via defined keyword search strings (Table 2), categorizing and filtering articles presenting relevant aspects for creating DT in civil infrastructure, emphasizing the technologies, tools, and framework involved in DT implementation. The string is based on Boolean operators that connect keywords

Table 1. Application of the PICOC criteria to civil infrastructure digital twin SLR

PICOC criteria	Definition in the SLR process
Population	Scientific journal productions on implementing and constructing digital twins in civil infrastructure assets.
Intervention	Read and select the papers that specify the development of digital twins for infrastructure assets.
Control	Journal papers with specific use of digital twins in great civil infrastructure works.
Outcomes	Report of the systematic literature review (SLR), including statistical analysis and synthesis of the most relevant articles.
Context	Application of the results for developments in future projects, establishing a methodology and definition in digital twins applied to civil engineering.

Table 2. Search strings for the SLR

Search string	Description
S1	(Bridge OR Tunnel) AND ("Digital Twin" OR "Virtual Model" OR "Virtual Twin" OR "Tunnel Information Modeling" OR "Bridge Information Modeling")
S2	(Infrastructure OR "Civil Infrastructure" OR "Civil Engineering") AND ("Digital Twin" OR "Virtual Model" OR "Virtual Twin" OR "Tunnel Information Modeling" OR "Bridge Information Modeling")

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to logically establish the search process. This process is an essential step for a systematic review and defines the quality of the results.

The word associations were structured to optimize the review of infrastructure applications within the civil engineering domain. String S1 was designed to restrict the search scope regarding the infrastructure type, focusing on the DT application for bridges and tunnels. String S2 was developed to encompass a broader range of civil infrastructure. Thus, it serves as a tool for establishing general methodologies applicable to civil infrastructure that could be employed in the infrastructure under analysis.

The authors opted to utilize references that did not explicitly address the creation of a digital twin but were linked to pertinent technologies. This decision followed a thorough analysis of articles related to SLR. Owing to the widespread use of the term "digital twin" in the scientific community, numerous studies have identified their scientific constructs as DT, even though they present a less intricate process or address fewer aspects of the digital twin definition than other studies involving related technologies do. Given the absence of a standardized definition for DT (Mahmoodian et al. 2022; Scully 2020), the search string S2, inclusion/exclusion criteria, and evaluation questions were proposed to encompass developments in correlated technologies relevant to the progression of the virtual twinning field.

To ensure the inclusion of a significant number of articles for analysis, the study incorporated terms that concerned virtual models, such as BIM platforms. The papers that presented these modeling strategies associated with technologies relevant to the DT context were included in the review process. The articles extracted from the chosen citation databases were filtered according to the inclusion and exclusion criteria stipulated to admit suitable papers for the review, as shown in Table 3.

Table 3. Criteria for paper selection

Type	Criteria
Inclusion	The paper was published in a journal after 2015.
	The study considers the use of digital twins for tunnels and bridges.
	The article presents technologies associated with the digital twin.
	The paper addresses issues related to digital twin applications.
Exclusion	The article is not written in English.
	The paper is a secondary or tertiary study.
	The article does not focus on a civil engineering infrastructure asset.
	The study does not answer the research questions.
	Only journal articles are considered. Conference papers are out of scope.
	Studies indexed in databases other than Scopus, Engineering Village, and Springer.
	The Universidade de São Paulo (Brazil) does not provide full access to the study content.
	The paper does not present a direct or integrated use of DT technology.

The inclusion and exclusion criteria established for paper selection are employed in the second main stage of the SLR. This step defines the articles selected for full reading and relevance evaluation. The execution and summarization phases are associated with applying the SLR protocol. The software StArt version 3.3 from the Laboratory of Research on Software Engineering of the Universidade Federal de São Carlos supported and optimized the execution process. The software generates an initial score on the basis of the paper information relevance, regarding the keywords established for the protocol search. Next, a reading priority is established, in which the title, abstract, and keywords of the articles are analyzed according to the inclusion and exclusion criteria.

The first batch of selected papers was subsequently read, and to evaluate and classify the studies in terms of their importance for the SLR, four evaluation questions (EQs) were elaborated (Table 4) on the basis of the research questions. For the score, the papers that presented a solution to the question EQi (EQi, where $i \in 1, 2, 3, 4$) received 1 point. Conversely, articles that did not completely answer the EQi received a zero value. Each paper was evaluated for each question, with the sum of the paper grades being the final score. Papers with a score lower than 2 (score < 2) were considered insufficiently relevant to answer the research questions and were therefore excluded from the review process. Table 5 displays the quality evaluation results of the 89 articles selected for summarization.

The final phase of the SLR, the summarization stage, encompasses the data extraction process of the relevant papers to develop a statistical analysis and data synthesis. The results extracted were organized and interpreted to translate the knowledge of the literature and answer the research questions. The summarized framework employed for the SLR is illustrated in Fig. 2. From the filtering process, the initial batch of articles, which presented more than 8,000 papers, was reduced to 89 relevant studies.

The SLR approach systematically searches a database, employing a forward/backward process for transparency, precision, replicability, and bias reduction (Allen et al. 2023). This methodology ensures accuracy in addressing research questions through careful planning, providing a publicly available framework for each step to enhance transparency and reproducibility. The analysis combines qualitative and quantitative assessments for increased reliability. However, similar to any other study, this study has several limitations. Owing to the scope restrictions imposed on the review protocol by the research context (see criteria for paper selection in Table 3), some crucial prior work may not be identified and included in the knowledge translation proposed by this paper. The main limitations of this review are related to the following:

- Gray literature in the form of books, Ph.D. and M.Sc. theses, conference papers, and technical reports were excluded from this review for feasibility purposes because of the copious amount of published data. As a result, some relevant and timely insights may have been missed.
- Only open-access journals and subscription-based journals in which Universidade de São Paulo provides full access to the

Table 4. Paper quality assessment questions

Questions to rank papers	Grade
EQ1: Does the research specify the life-cycle phase in which the digital twin is deployed?	Yes - value 1; No - value 0
EQ2: Does the article indicate the purpose of employing a digital twin?	
EQ3: Does the research present the technologies associated with digital twins?	
EQ4: Does the study present challenges, advancements, and future opportunities related to adopting digital twins for civil engineering infrastructure?	

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Table 5. Article classification scores

References	Score
Kyvelou et al. (2022), Dang et al. (2022), Wu et al. (2022c), Bono et al. (2022), and Gürdür Broo et al. (2022) Ghahari et al. (2022), Machado and Futai (2024), Kim and Kim (2020), and McGuire et al. (2016) Wu et al. (2022b), Futai et al. (2022), Yu et al. (2022), Omer et al. (2021), and Mohammadi et al. (2021) Kaewunruen et al. (2021), Nili et al. (2021), Marra et al. (2021), Dan et al. (2022), and Ye et al. (2020) Xu and Turkan (2020), Lu and Brilakis (2019), Rashidi and Karan (2018), and Jeong et al. (2017) Chakraborty and Adhikari (2021), Chen and Whyte (2022), and Taraben and Morgenthal (2021) Meixedo et al. (2021), Li et al. (2022), Zou et al. (2019), Kwon et al. (2021), and Shen et al. (2022) Lei et al. (2022), Jiang and Jiang (2022), Kusimba et al. (2022), Wu et al. (2022a), and Zhao et al. (2023) Talebi et al. (2023), Jasiński et al. (2023), Mohammadi et al. (2023), Naraniecki et al. (2023), and Jäkel et al. (2023) Nhamage et al. (2023), Chacón et al. (2023), Hielscher et al. (2023), Yu et al. (2023), and Ye et al. (2023) Fleming et al. (2016), Adibfar and Costin (2022), Wang et al. (2023), and Chacón et al. (2025) Mandirola et al. (2022), Scianna et al. (2022), Yu et al. (2022), Zhao et al. (2022), and Samadi et al. (2021) Jiang et al. (2021), Levoni et al. (2021), Lin et al. (2021), Sharafat et al. (2021), and Patel et al. (2021) Yu et al. (2020), Omer et al. (2019), Shim et al. (2019), and Koch et al. (2017)	4
Lu and Brilakis (2020), Meixedo et al. (2022), Kim et al. (2018), Merkin et al. (2021), and Yoon et al. (2022) Baisthakur and Chakraborty (2021), Jeong et al. (2016), Febrianto et al. (2022), and Nicoletti et al. (2023) Mafipour et al. (2023), Nhamage (2023), Catibac (2023), Zhang et al. (2024), and He et al. (2024) Kang et al. (2021), Han et al. (2020), Hu et al. (2023), Torzoni et al. (2024), and Lin et al. (2023)	3
Sun and Liu (2022), Brucherseifer et al. (2021), Sacks et al. (2020), and Monsberger et al. (2022) Guo and Fang (2024) and Chacón et al. (2024)	2

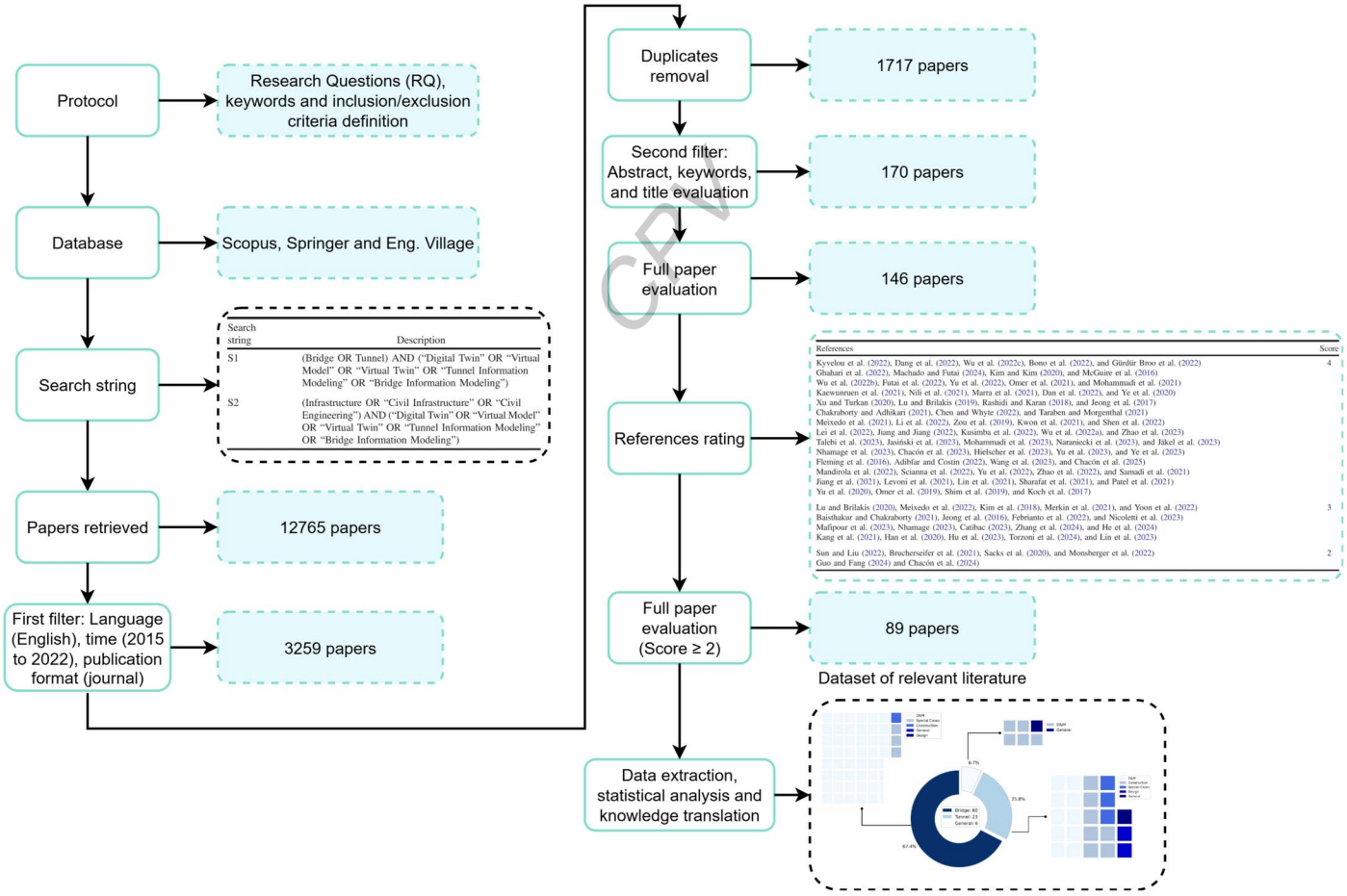


Fig. 2. SLR methodology.

content were considered for the SLR. This access constraint may have introduced selection bias.

- The selection criteria were limited to primary, original journal papers authored in English, which may have led to the underrepresentation of region-specific developments or research published in other languages.
- The research database was restricted to studies indexed in Scopus, Engineering Village, and Springer. While these are

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comprehensive sources, relevant studies indexed exclusively in other databases may not have been captured.
- Practitioner-focused literature (e.g., white papers, industry reports, and trade magazines) may have been omitted due to the academic nature of the search databases. This may have resulted in the exclusion of some practical insights.

Results and Discussion

This section presents the findings from the literature review process described in the section "Methodology." The relevant articles obtained by the SLR were subjected to a rigorous synthesis and compilation of the key findings. The first step focused on exploratory data analysis, which quantifies the papers selected and the evolution of the research field over time. The findings were subsequently divided into the main topics that help to answer the proposed research questions: the life-cycle phase, intended uses, level of development, and challenges/future directions.

Exploratory Data Analysis

Fig. 3 illustrates the annual distribution of journal papers published from 2015 to October 2024 in English, revealing an upward trend in adopting digital twins and their associated technologies for civil infrastructure. The data were obtained through a search within one of the main journal indexers (Elsevier) using the string ["Digital Twin" AND ("Civil Infrastructure" OR "Civil Engineering" OR Infrastructure)]. Different search string search configurations were tested, all indicating significant growth in interest in research in this area. This substantial increase in research output indicates the rising popularity of digital twins among the scientific community and their increasing consideration as a viable solution for the challenges faced in the field of study (Adeagbo et al. 2024).

RQ1: Life-Cycle Stages

RQ1 aims to identify the life-cycle phases in which digital twins are employed. The life-cycle stages describe a critical phase for the infrastructure life, from its design to the decommissioning phase. Thus, establishing digital twin adoption for each stage supports understanding the technology most commonly used and the particularities for different life-cycle periods of the asset. The life-cycle phases established by the SLR were divided into design, construction, operation and maintenance (O&M), special cases, and general cases. Articles were classified as "special cases" when the application of DT was focused on managing atypical conditions, and natural hazards [e.g., seismic events (Ghahari et al. 2022; Lin et al. 2021), extreme conditions (Kaewunruen et al. 2021), fires (Wu et al. 2022b), and emergency evacuation measures (Wu et al. 2022c; Han et al. 2020)]. On the other hand, "general cases" refers to articles that either cover multiple life-cycle phases or propose broadly applicable DT frameworks that are not restricted to a single phase or specific event.

The studies were initially organized according to Table 6, classifying each reference by the life-cycle stage approached and the type of infrastructure used for the development of DT.

Fig. 4 shows the data analysis of the life-cycle stage to which digital twin technology was applied. It illustrates the significant prevalence of the application to the domain of O&M of infrastructure, accounting for 75.3% of the cases, followed by processes focused on the asset construction stage at 11.2%.

Another key aspect of the analysis focuses on the frequency of investments in each category of infrastructure asset. RQ1 further aims to examine the specific application of the technology to bridges and tunnels. In Fig. 5 the distribution of DT developments in bridges, tunnels, and other broader applications in illustrated. The term "general" refers to instances in which the technology is employed for large-scale infrastructure, encompassing bridges and tunnels.

DT is most commonly applied to bridges (67.4%), followed by tunnels (25.8%) and general applications (6.7%). The lower number of tunnel applications compared with those of bridges can be attributed to the difficulties in monitoring underground infrastructure. Tunnel monitoring often involves additional challenges, such as a lack of natural light, limited access, and the need for specific sensing technologies, whereas monitoring bridges may be more

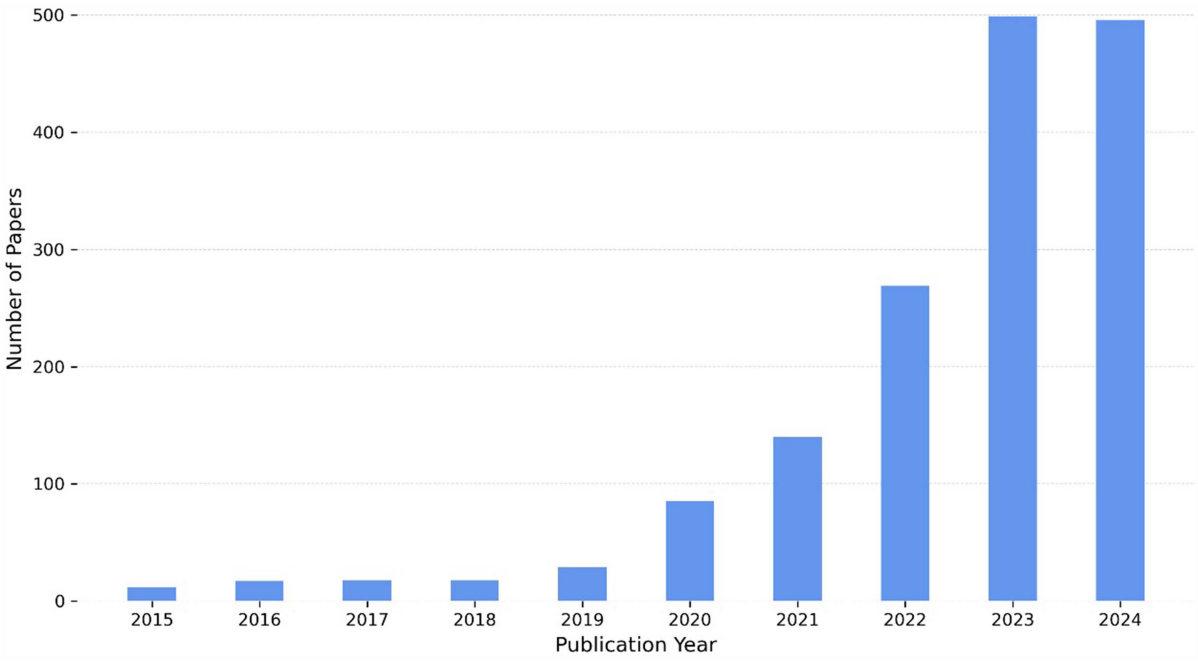


Fig. 3. Application of digital twins to civil infrastructure.

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Table 6. Life-cycle stage classification

Asset	Life-cycle stage	Reference
Bridges	O&M	Dang et al. (2022), Bono et al. (2022), and Gürdür Broo et al. (2022) Talebi et al. (2023), Mandirola et al. (2022), and Scianna et al. (2022) Yu et al. (2022), Jasiński et al. (2023), and Kang et al. (2021) Zhao et al. (2022), Samadi et al. (2021), and Mohammadi et al. (2023) Omer et al. (2021), Mohammadi et al. (2021), and Nicoletti et al. (2023) Jiang et al. (2021), Nili et al. (2021), and Marra et al. (2021) Baisthakur and Chakraborty (2021) and Lu and Brilakis (2020) Dan et al. (2022), Patel et al. (2021), and Ye et al. (2020) Omer et al. (2019) and Shim et al. (2019) Rashidi and Karan (2018), Jeong et al. (2017), and Jeong et al. (2016) McGuire et al. (2016), Meixedo et al. (2022), and Kim et al. (2018) Naraniecki et al. (2023), Wang et al. (2023), and Xu and Turkan (2020) Meixedo et al. (2021), Li et al. (2022), and Kwon et al. (2021) Yoon et al. (2022), Adibfar and Costin (2022), and Lei et al. (2022) Febrianto et al. (2022), Zhao et al. (2023), and Lu and Brilakis (2019) Chacón et al. (2023), Lin et al. (2023), and Hu et al. (2023) Mafipour et al. (2023), Nhamage et al. (2023), and Jäkel et al. (2023) Hielscher et al. (2023), Nhamage (2023), and Yu et al. (2023)
	Construction	Kyvelou et al. (2022) and Zou et al. (2019)
	Special cases	Ghahari et al. (2022), Lin et al. (2021), and Kaewunruen et al. (2021)
	General	Futai et al. (2022) and Chacón et al. (2025)
	Design	Guo and Fang (2024)
Tunnels	O&M	Yu et al. (2021), Levoni et al. (2021), Kim and Kim (2020), and Catibac (2023) Yu et al. (2020), Merkin et al. (2021), Wu et al. (2022a), and Zhang et al. (2024) Machado and Futai (2024) and He et al. (2024)
	Construction	Koch et al. (2017), Fleming et al. (2016), and Monsberger et al. (2022) Jiang and Jiang (2022), Kusimba et al. (2022), and Torzoni et al. (2024) Ye et al. (2023)
	Special cases	Wu et al. (2022c), Wu et al. (2022b), and Han et al. (2020)
	General	Chen and Whyte (2022)
	Design	Sharafat et al. (2021) and Shen et al. (2022)
General	O&M	Sun and Liu (2022) and Brucherseifer et al. (2021) Chakraborty and Adhikari (2021) and Chacón et al. (2024)
	Construction	Taraben and Morgenthal (2021) and Sacks et al. (2020)

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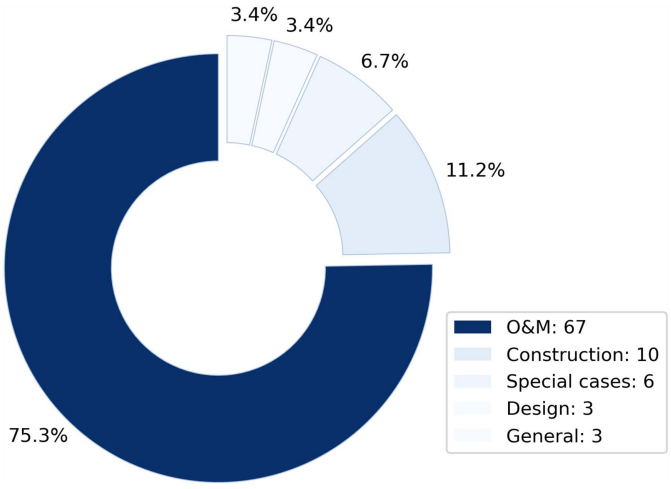


Fig. 4. Life-cycle phase of infrastructure DT.

accessible and straightforward because of their visible structure and easier access (Bhalla et al. 2005). The Federal Highway Administration's National Bridge Inventory, which has tracked more than 621,000 bridges since 1995 (FHWA 1995), contrasts with the National Tunnel Inventory, which has recorded data on just over 560 tunnels since 2015. Consequently, there is little historical data available for tunnels in digital twin applications during the O&M phase. As a result, tunnels exhibit a more evenly distributed frequency of applications across the life-cycle phases, whereas bridges show a notably higher frequency in the O&M phase. Advancements in monitoring and inspection technologies could help overcome these challenges.

RQ2: Intended Uses

RQ2 aims to identify the specific purposes underlying the implementation of digital twins in civil infrastructure such as bridges and tunnels. Unlike the categorization based on the life-cycle stage and asset type, which is relatively straightforward, the classification that pertains to the digital twin's intended use involves a certain level of subjectivity. The intended use was classified according to Scully (2020) to establish an application metric that correlates the classification process to the digital twinning complexity level: digitize, visualize, simulate, emulate, extract, orchestrate, and predict. This classification seeks to understand the complexity and maturity level of the methodology for civil infrastructure.

A digital twin focused on simulation intends to accurately represent the physical system behavior under real-life conditions within its environment. For this development stage, the FEM and BIM/BrIM are typically employed to implement a digital model. A digital twin for extraction focuses on establishing data streams from the physical asset to the virtual system. Cloud-based (Dang et al. 2022) and online (Meixedo et al. 2022) strategies have emerged as

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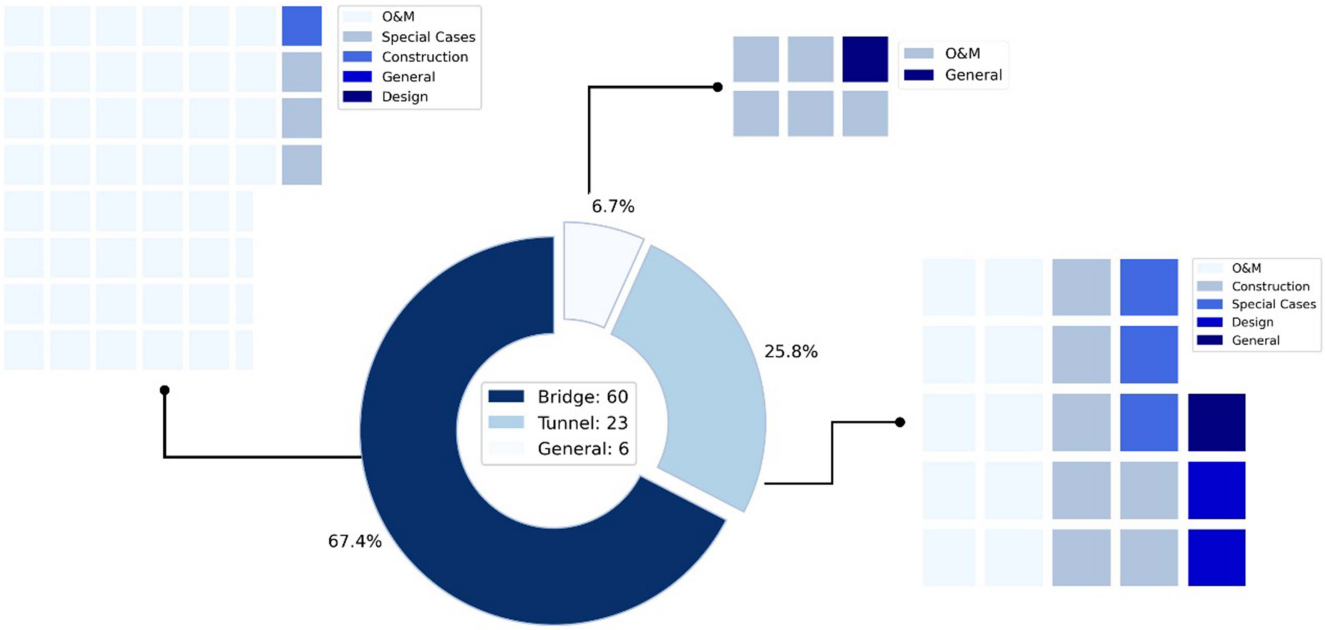


Fig. 5. DT frequency for civil infrastructure.

solutions for integrating structural health monitoring (SHM) data into the virtual model and providing continuous infrastructure condition assessment. These online frameworks have been associated with deep learning algorithms to perform structural damage detection. A DT for orchestration aims at asset management and updating, focusing on decision-making. Thus, a model with this complexity level can incorporate an O&M decision scheme by associating long-term SHM and a statistical finite-element method for synthesizing data and physics-based models, achieving a DT of a self-sensing structure (Febrianto et al. 2022). A predictive digital twin presents the highest complexity level observed, allowing us to foresee future infrastructure behavior (Yu et al. 2022; Kang et al. 2021; Jiang et al. 2021). Table 7 summarizes the intended uses of digital twins for civil engineering infrastructure.

Fig. 6 displays the correlation between the life-cycle phase and the intended use of DTs. Notably, simulation and extraction denote a significant portion of O&M and construction applications, stages in which reproducing the physical system behavior under real-life

Table 7. Objective standardization

Asset	Objective/use	Reference
Bridge	Visualize	Mandirola et al. (2022), Samadi et al. (2021), and Mohammadi et al. (2021, 2023) Marra et al. (2021), Lu and Brilakis (2019), Rashidi and Karan (2018), and McGuire et al. (2016) Omer et al. (2021), Lu and Brilakis (2020), Hu et al. (2023), and Mafipour et al. (2023)
	Simulate	Kyvelou et al. (2022), Bono et al. (2022), Ghahari et al. (2022), and Zhao et al. (2022) Baisthakur and Chakraborty (2021), Lin et al. (2021), Kaewunruen et al. (2021), and Nili et al. (2021) Dan et al. (2022), Patel et al. (2021), Ye et al. (2020), Xu and Turkan (2020), and Nhamage et al. (2023) Omer et al. (2019), Shim et al. (2019), and Zou et al. (2019) Adibfar and Costin (2022), Kusimba et al. (2022), Talebi et al. (2023), and Jasiński et al. (2023) Nhamage (2023) and Nicoletti et al. (2023)
	Extract	Dang et al. (2022), Gürdür Broo et al. (2022), Mandirola et al. (2022), and Futai et al. (2022) Jeong et al. (2016, 2017) and Meixedo et al. (2021, 2022) Li et al. (2022), Lei et al. (2022), Zhao et al. (2023), Kim et al. (2018), and Wang et al. (2023) Naraniecki et al. (2023), Guo and Fang (2024), Lin et al. (2023), and Chacón et al. (2024)
	Orchestrate	Febrianto et al. (2022), Jäkel et al. (2023), and Chacón et al. (2023)
	Predict	Yu et al. (2022), Kang et al. (2021), Jiang et al. (2021), Kwon et al. (2021), and Yoon et al. (2022) Hielscher et al. (2023) and Yu et al. (2023)
Tunnels	Visualize	Fleming et al. (2016) and Catibac (2023)
	Simulate	Han et al. (2020), Shen et al. (2022), Koch et al. (2017), Zhang et al. (2024), and He et al. (2024)
	Extract	Wu et al. (2022b, c), Levoni et al. (2021), and Monsberger et al. (2022) Merkin et al. (2021), Jiang and Jiang (2022), Wu et al. (2022a), and Machado and Futai (2024)
	Orchestrate	Yu et al. (2021), Sharafat et al. (2021), and Torzoni et al. (2024)
	Predict	Kim and Kim (2020), Yu et al. (2020), Chen and Whyte (2022), and Ye et al. (2023)
General	Visualize	Taraben and Morgenthal (2021)
	Simulate	Sun and Liu (2022), Brucherseifer et al. (2021), and Sacks et al. (2020)
	Extract	Chakraborty and Adhikari (2021) and Chacón et al. (2025)

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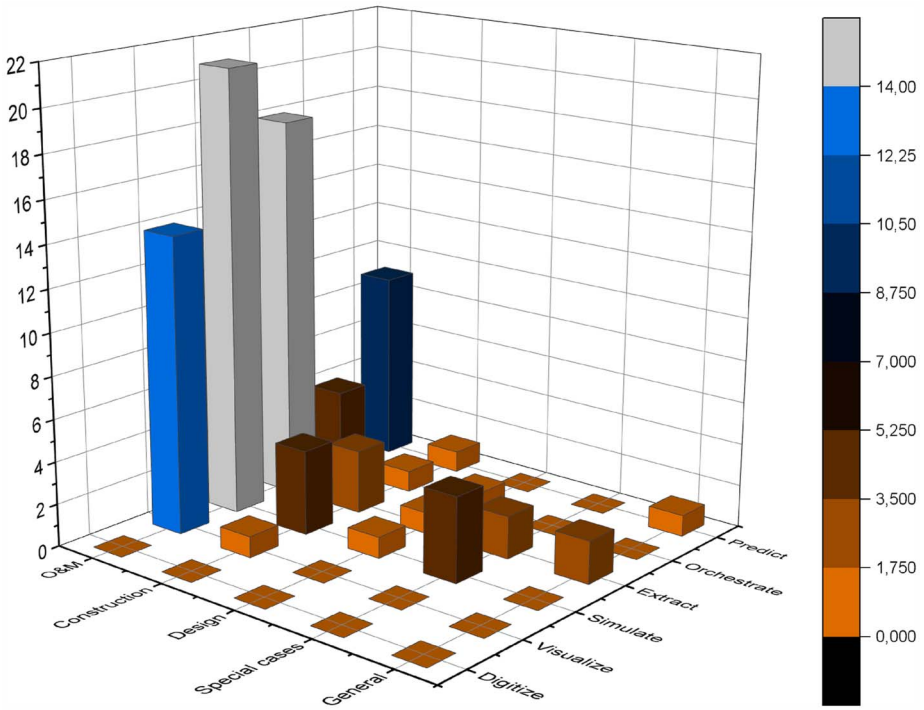


Fig. 6. Connection between life-cycle stage and use.

conditions and assessing data extracted from the physical asset is essential to define the approaches that support the development of constructive and operational activities.

The review results confirm that classification of the intended use of digital twins involves varying levels of complexity: digitizing, visualizing, simulating, emulating, extracting, orchestrating, and predicting (Scully 2020). Each objective represents an evolution in DT development, adding capabilities to the virtual model. Examples of uses include digitizing data, creating digital replicas, accurate simulation, emulating physical systems, real-time data extraction, virtual model orchestration, and future behavior prediction. Tunnels tend to concentrate at higher complexity levels, especially in field data extraction, whereas bridges and large infrastructure projects generally have more simulation use cases. The most common DT applications include simulation, extraction, and visualization, primarily for infrastructure management during the O&M life-cycle stage. Simulation is a key application in both the O&M and construction stages, where replicating physical system behavior under real-life conditions is essential.

RQ3: Associated Technologies and Level of Development

RQ3 maps the leading technologies implemented for a civil infrastructure DT, which are categorized into various aspects, including modeling, SHM (Tan et al. 2022), numerical analysis, code development (Naser 2023), inspection, machine learning (Ruiz et al. 2024), deep learning (Yang 2022; Ruiz et al. 2025; Rajkumar et al. 2023), databases, virtual reality/augmented reality (VR/AR) (Awadallah and Sadhu 2023), and single-board computers.

Fig. 7 shows the frequency of application of leading technologies in the construction of digital twins. The most common technologies applied to civil engineering digital twins are modeling, structural sensing, and numerical analysis, which are linked through information exchanges facilitated by the development of programming routines. Technologies used for field data acquisition, such as IoT, image capture, sensors, and machine learning, require synchronization with DT models to ensure effective implementation. The low utilization of virtual reality may indicate a potential future direction for its integration with civil engineering digital twins, mainly because of its benefits for operation and maintenance activities.

The critical aspect lies in the integration of various technologies into the DT process, which is also an objective of RQ3. The levels of DT development are closely tied to the incorporation of new technologies into the twinning process, with different methodologies and technologies being applied for various purposes. The use of only one technology is rare in creating a virtual copy of reality. Based on the data collected, the association of at least three of these technologies is a limiting factor in the methodology for constructing digital twins for analytical assets. Fig. 8 shows the association of technologies with DTs. The application of more than three of the standardized technologies represents more than 66% of the applications.

In addition, the connection between the life-cycle phases and technologies is illustrated qualitatively in Fig. 9.

In the design phase, virtualization relies on high-level modeling with detailed structural calculations fed by field data. These are then analyzed using visual immersion technologies and algorithms for interpretation. Modeling and numerical analysis are required during construction, with a focus on chronological construction steps (simulation). Additionally, there is a certain level of data acquisition automation, which can be based on the Internet of Things (IoT) technology and data archiving during construction. These measures are essential to address unforeseen issues during asset implementation.

The O&M management phase is associated with obtaining and connecting data on structural behavior and performance, a crucial stage for applying the digital twin methodology to civil infrastructure. This phase involves the most comprehensive application of technologies and processes, creating a holistic digital representation

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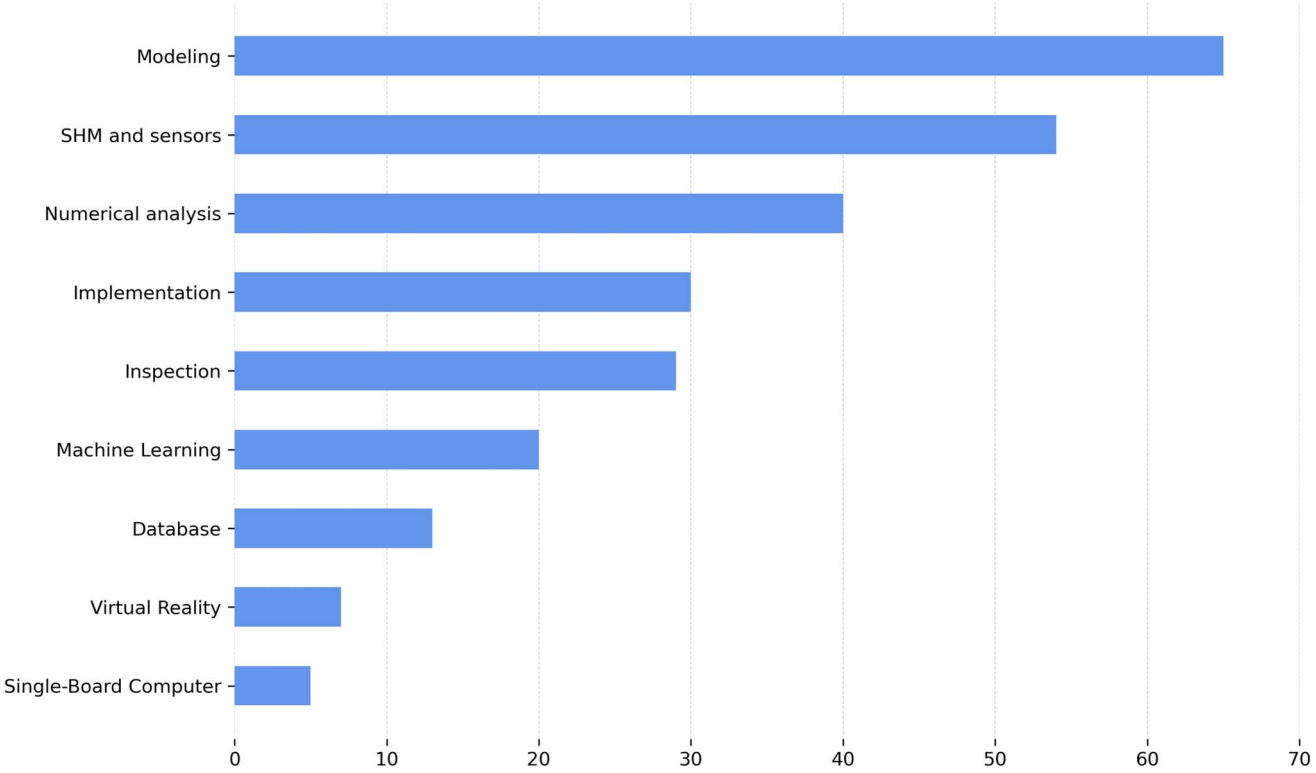


Fig. 7. Application frequency of technologies in infrastructure digital twin.

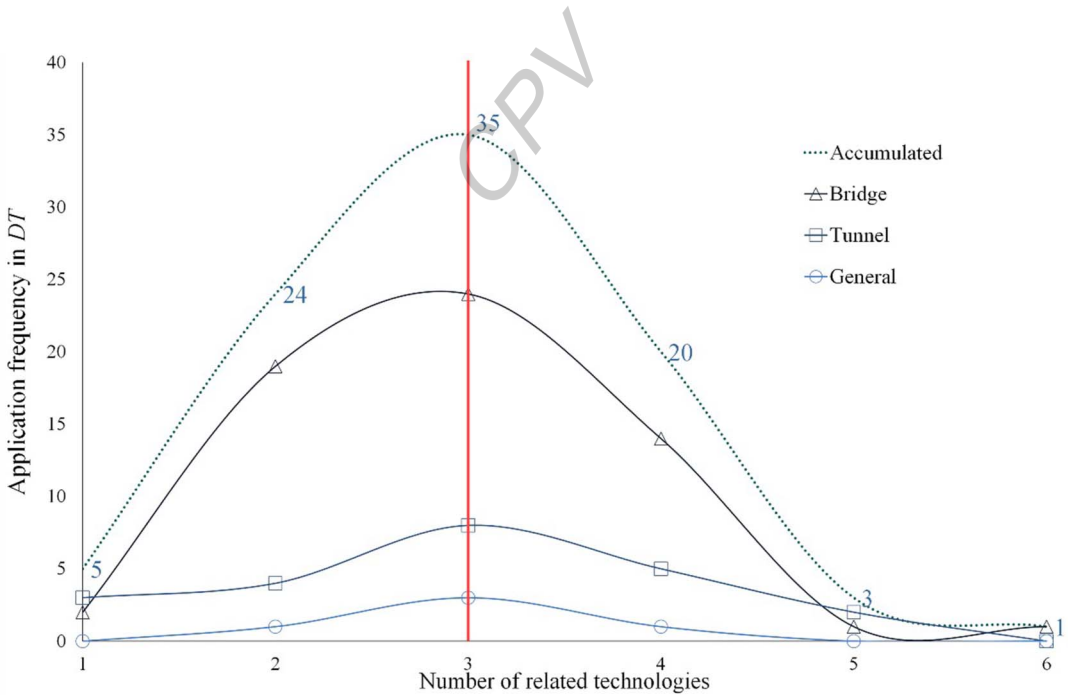


Fig. 8. Technologies association in infrastructure digital twin.

of the structure. Structural health monitoring, alert triggers, data processing using artificial intelligence (AI) methodologies, and augmented reality resources are studied to aid investment decision-making throughout the structure life cycle. O&M is also the most extended phase in the structure life cycle. Therefore, data acquisition is crucial and requires tools to establish a specific knowledge base for the virtualized structure.

The current rapid development of AI tools plays a key role in advancing DT systems by enabling predictive maintenance and proactive decision-making. Machine learning models trained on historical and real-time data can detect structural anomalies and forecast deterioration, enhancing infrastructure reliability and safety (Kang et al. 2021; Nili et al. 2021). Applications such as computer vision support automated damage detection (Kwon et al. 2021), while edge

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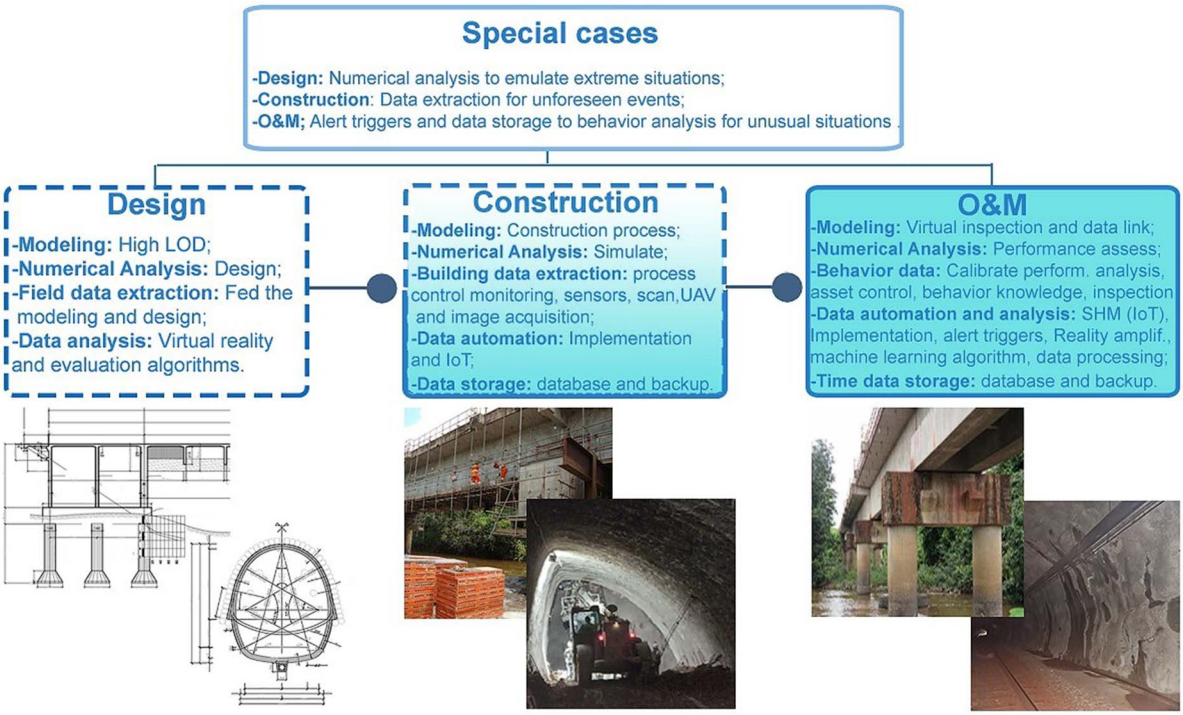


Fig. 9. Life-cycle phase of infrastructure DT. (Images by authors.)

computing and cloud-based architectures facilitate real-time data processing for critical assets (Yu et al. 2020). DTs also contribute to fatigue performance analysis and the estimation of structural component lifespan (Jiang et al. 2021; Yu et al. 2020). However, the effectiveness of these AI-driven capabilities depends heavily on robust data acquisition and management, highlighting the importance of strong data governance within infrastructure organizations.

Notably, both the bridge and tunnel life-cycle stages are characterized by intensive monitoring activities. Therefore, a substantial share of studies have associated SHM (Gürdür Broo et al. 2022), embedded sensors (Baisthakur and Chakraborty 2021), IoTs (Scianna et al. 2022), and remote monitoring techniques (Ye et al. 2020) with digital twins. One of the central aspects of a DT methodology is the exchange of information between the real and virtual worlds. The interaction and convergence among previously acquired data and those continuously provided by monitoring sensors bolster the closed data ecosystem and enable the updated DT to represent the actual asset, supporting decision-making activities appropriately. Therefore, the life-cycle stages characterized by a more significant implementation of SHM activities have garnered substantial research attention, particularly concerning the current focus on DT technologies. Given that monitoring campaigns are pivotal O&M activities, this phase has yielded the most pronounced outcomes, as previously illustrated in Fig. 4.

Using digital twins to simulate behavior in scenarios involving special situations during different life-cycle phases serves significant purposes. First, it facilitates the emulation of extreme requests during the design phase, aiding in understanding how the structure will respond to such conditions. Second, it enables the extraction and storage of information from unexpected design issues, primarily focused on optimizing construction decisions. Finally, digital twins can be employed to develop an alert system for readings that deviate from the predicted behavior in the design phase and are observed during construction, allowing prompt action. Additionally, the historical storage of occasional events in the structure provides valuable data for future analyses and decision-making processes. By combining these functionalities, digital twins play a crucial role in enhancing the overall performance and resilience of infrastructure throughout its life cycle.

The transition of digital twin technology from conceptual frameworks to real-world infrastructure applications requires a structured and strategic approach. This involves clearly defined key performance objectives and the development of a robust data ecosystem capable of integrating heterogeneous sources. Key enablers include scalable storage solutions, data quality protocols, data interoperability, and multidisciplinary collaboration across engineering, data science, and software development. Although many studies remain conceptual, recent studies have demonstrated practical applications in bridge and tunnel infrastructure. For instance, Meixedo et al. (2022) developed a DT framework that integrates SHM and three-dimensional modeling for a historical bridge, enabling real-time visualization and predictive maintenance. Febrianto et al. (2022) proposed a hybrid system combining SHM and BIM to support bridge asset management. In the tunnel infrastructure, Yu et al. (2023) introduced a DT-based risk evaluation system utilizing real-time sensor inputs for emergency response, while He et al. (2024) implemented a simulation-based DT to model tunnel fire scenarios. These examples reflect growing adoption through modular and pilot-scale implementations. However, broader applications are still challenged by integration complexity, organizational readiness, and costs. Nonetheless, these cases confirm that DTs and their associate technologies are actively shaping modern infrastructure management rather than remaining theoretical constructs.

In this context, emphasizing incremental and modular implementation strategies can make digital twins more accessible to industry professionals. Rather than requiring a full-scale digital transformation, practical adoption can begin with isolated components tailored to stakeholder needs or based on deployments with lower maturity levels. Aligning DT tools with existing asset management systems,

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leveraging open standards for data interoperability, and focusing on specific use cases help reduce complexity and implementation costs. In industrial settings, an important focus should be the integration of DT workflows into company culture. This involves fostering collaboration among multidisciplinary stakeholders and promoting standardized data integration practices across the sectors to ensure that DT solutions are aligned with field conditions, operational objectives, and the available technical capacity.

RQ4: Main Challenges and Future Directions

DT development for civil bridges and tunnels is still incipient, and the current practices do not provide the necessary skills for adequately developing and implementing the technology. Additionally, there are no well-defined standards, processes, or tools to support their development Thus, projects concerning digital twins are multidisciplinary efforts of civil engineers and computer scientists, which combine practices, procedures, terminology, the level of maturity of the employed technology, and data fluency of both domains and from different stakeholders (Gürdür Broo et al. 2022).

For the specific context of bridges and tunnels, some of the main challenges and innovations for DTs are related to organizing and storing all the structural information necessary to compare with real-time SHM data, correlating real-life data with computational simulations, improving inspection and maintenance activities, automating alerts through machine learning and new technologies, incorporating sensors and inspection information into the DT for continuous monitoring and diagnostics, predicting future structural behavior based on the stored data accumulated over the structure life cycle, applying computer vision to detect and track damage evolution, combining cloud-based computational analysis with real data, and enabling comprehensive infrastructure management by integrating inspection, monitoring, and maintenance with automatic analysis into the DT (Futai et al. 2022). Integrating data from real assets to the virtual model is one of the main concerns stipulated as essential future improvements for digital twin research (Marra et al. 2021; McGuire et al. 2016; Chakraborty and Adhikari 2021; Zou et al. 2019).

Managing and processing data, especially in civil infrastructure, is a significant challenge since data in civil infrastructure projects are often presented in a disconnected and heterogeneous manner. Understanding the relationship among various data parameters is also a challenge for developing DT structures. Tunnel monitoring often involves additional challenges, such as a lack of natural light, limited access, and the need for specific sensing technologies.

Digital twins depend on cutting-edge data processing solutions, such as machine learning and optimization algorithms (Sacks et al. 2020). Thus, future research may focus on overcoming the current constraints of these techniques. At their core, AI-based solutions rely on a few key factors, such as generalization, data availability, historical context, training time, and computational capacity. Generalization is crucial because it enables models to apply their knowledge to new, unseen data, making them versatile for practical applications. Thus, unsupervised and semisupervised learning algorithms should be used to increase generalization (Dang et al. 2022).

In addition to technical knowledge regarding asset structural and operational behavior, for implementing DTs it is essential to assimilate expertise from new subjects, such as project management, data management, data quality, data architecture, cyber security, and even privacy and ethics. Integrating these different areas and providing comprehensive communication between stakeholders are vital to bridging the gap between the real world and the virtual world, enabling the implementation of a full-scale digital twin. In addition to the organizational barrier, implementing DT will require overcoming technical, sociological, and commercial challenges (Sacks et al. 2020).

Another important point for future development is related to the DT nomenclature. Many studies often develop high complexity level models associated with current technologies; however, they are unaware that they are creating a twinning of the infrastructure. The term DT and its framework are not widespread, promoting problems with increasing the number of virtual twinning applications. Approximately 76% of review papers use the term DT to describe the application methodology, whereas others build complex models using more current technologies without acknowledging that they are creating DTs. Therefore, a standardization of the definition of the digital twin is essential to determine what differentiates a digital twin from a less complete virtual model.

Moreover, future state-of-the-art reviews may benefit from expanding the scope of analysis of this study to include gray literature, such as theses, conference proceedings, and technical reports. Additionally, incorporating practitioner-oriented sources, such as industry reports and trade publications, may enrich the understanding of practical applications and real-world challenges. Broadening the range of databases enhances the comprehensiveness of the review and reduces potential publication bias. Finally, considering publications in languages other than English is recommended for a more inclusive and globally representative synthesis of the literature.

In light of these considerations, future research on DT in civil infrastructure, particularly for bridges and tunnels, could benefit from addressing specific research questions. These questions are proposed to guide complementary systematic reviews and further investigations:

- What are effective strategies for integrating real-time sensor and inspection data into DT models for continuous and automated monitoring?
- Which data architectures best support the management of disconnected and heterogeneous civil infrastructure data for DT functionality?
- How can infrastructure administrators ensure secure, efficient data acquisition and management to meet the demands of DT systems?
- How can DT system design overcome monitoring constraints in civil infrastructure (e.g., poor lighting, limited access)?

Conclusion

This systematic literature review examined the advancements in DT technology for bridges and tunnels, aiming to address four research questions related to life-cycle phases, intended uses, associated technologies, and main challenges and perspectives. One notable challenge encountered was the inconsistent use of the term "digital twin," which complicates the identification of virtual twinning applications within the published literature. To mitigate this problem, this research employed alternative search strings to capture nonreferenced DTs. The findings revealed that digital twins have been adopted in all stages of the civil infrastructure life cycle, with a predominant focus on O&M, representing 75.3% of the cases. Bridges are the most common application of DTs, followed by tunnels and general applications. The varying levels of complexity in the intended use of digital twins are illustrated, ranging from digitizing and visualizing to more advanced functions such as simulating, emulating, extracting, orchestrating, and predicting. Tunnels tend to have a higher concentration of virtual twin cases at the more complex levels, particularly in field data extraction, while bridges

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and large infrastructures have more simulation cases. The leading technologies used in civil infrastructure DTs include modeling, SHM, numerical analysis, code development, inspection, machine learning, databases, VR/AR, and single-board computers. The integration of multiple technologies is common in DT construction, with at least three technologies being associated with more than 66% of applications. The most common technologies applied are modeling, structural sensing, and numerical analysis. Significant challenges in implementing DTs stem from the complexity of managing and processing data, given the heterogeneous nature of civil infrastructures, as well as the difficulty in maintaining a continuous connection between real and virtual assets. Some studies also overlook organizational challenges in implementing DTs. The construction of digital twins presents considerable opportunities and benefits for large-scale civil infrastructure. However, adapting to the large scale and complexities inherent in civil engineering, particularly for assets such as bridges and tunnels, remains an important consideration for future advancements.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Author Contributions

Lucas Bellini Machado: Conceptualization; Data curation; Formal analysis; Methodology; Validation; Writing – original draft; Writing – review and editing. Ruan Richelly Santos: Conceptualization; Formal analysis; Investigation; Methodology; Writing – original draft; Writing – review and editing. Dianelys Vega Ruiz: Conceptualization; Formal analysis; Investigation; Methodology; Writing – original draft; Writing – review and editing. Túlio Nogueira Bittencourt: Project administration; Supervision. Marcos Massao Futai: Funding acquisition; Methodology; Project administration; Supervision.

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